



The Coronavirus Disease of 2019

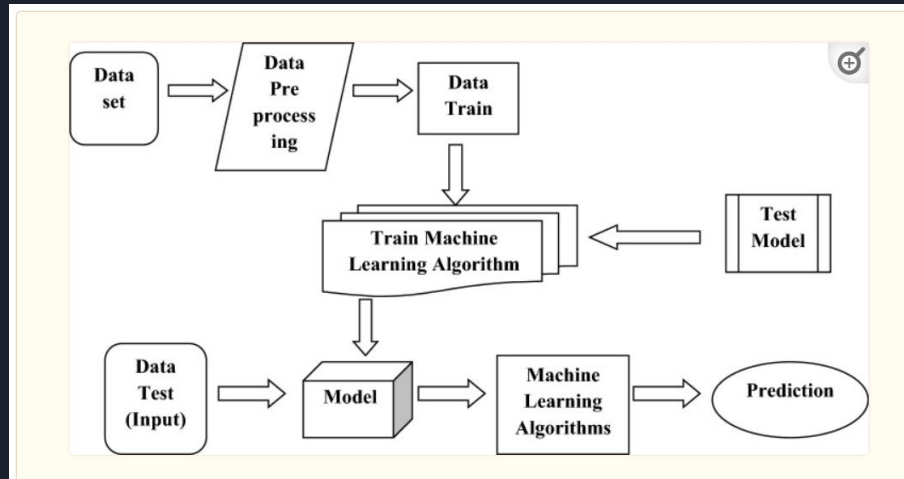
Hafsah Ashraf

Erin Hemsley

Introduction

In December 2019, Wuhan, China, reported an outbreak of a highly infectious respiratory disease that was later identified as Coronavirus Disease 2019. COVID-19 has since spread throughout the world, affecting many people's lives in various ways.

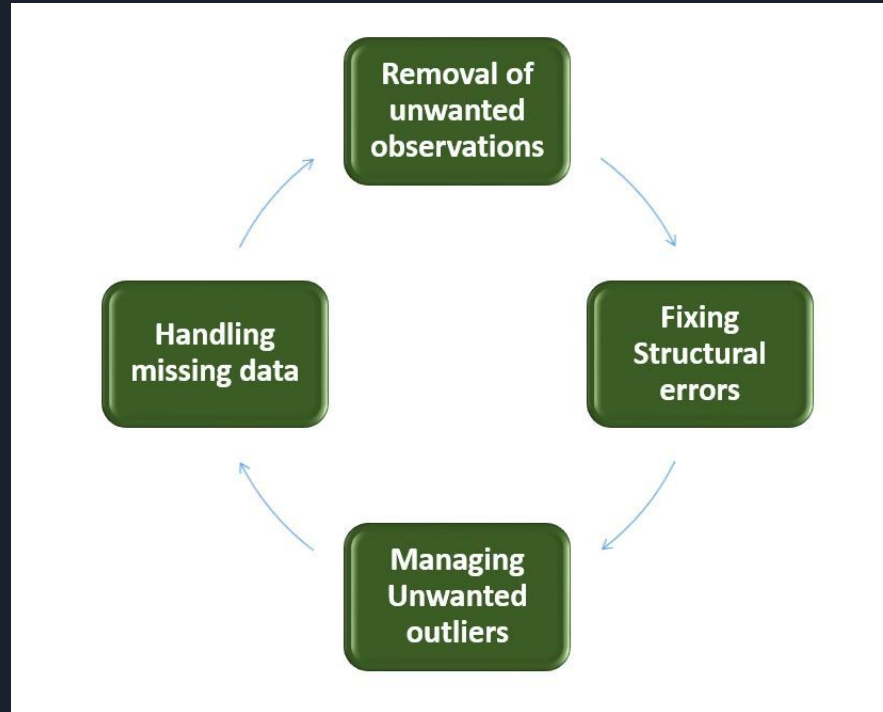
In times of crisis, the effectiveness of policymakers' decision-making is heavily reliant on their ability to integrate and interpret the information. The COVID-19 crisis forces governments to make difficult decisions in the name of public health and safety. To combat the COVID-19 pandemic, we have developed a machine learning model to help predict outbreaks, allowing the medical system and government to prepare and allocate resources ahead of time.



Dataset



Data Cleansing



Independent Variables

Variable	Description
total_vaccinations	Total number of COVID-19 vaccination doses administered
people_vaccinated	Total number of people who received at least one vaccine dose
people_fully_vaccinated	Total number of people who received all doses prescribed by the initial vaccination protocol
total_boosters	Total number of COVID-19 vaccination booster doses administered (doses administered beyond the number prescribed by the vaccination protocol)
new_vaccinations	New COVID-19 vaccination doses administered (only calculated for consecutive days)
new_vaccinations_smoothed	New COVID-19 vaccination doses administered (7-day smoothed). For countries that don't report vaccination data on a daily basis, we assume that vaccination changed equally on a daily basis over any periods in which no data was reported. This produces a complete series of daily figures, which is then averaged over a rolling 7-day window
total_vaccinations_per_hundred	Total number of COVID-19 vaccination doses administered per 100 people in the total population

Independent Variables

population	Population (latest available values). See https://github.com/owid/covid-19-data/blob/master/scripts/input/un/population_latest.csv for full list of sources
population_density	Number of people divided by land area, measured in square kilometers, most recent year available
median_age	Median age of the population, UN projection for 2020
aged_65_older	Share of the population that is 65 years and older, most recent year available
aged_70_older	Share of the population that is 70 years and older in 2015
gdp_per_capita	Gross domestic product at purchasing power parity (constant 2011 international dollars), most recent year available
extreme_poverty	Share of the population living in extreme poverty, most recent year available since 2010
cardiovasc_death_rate	Death rate from cardiovascular disease in 2017 (annual number of deaths per 100,000 people)
diabetes_prevalence	Diabetes prevalence (% of population aged 20 to 79) in 2017
female_smokers	Share of women who smoke, most recent year available
male_smokers	Share of men who smoke, most recent year available



Multiple Linear Regression Models

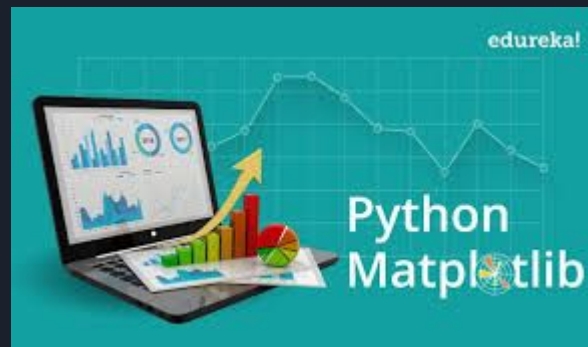
Multiple linear regression formula

The formula for a multiple linear regression is:

$$y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \epsilon$$

- y = the predicted value of the dependent variable
- B_0 = the y-intercept (value of y when all other parameters are set to 0)
- $B_1 X_1$ = the regression coefficient (B_1) of the first independent variable (X_1) (a.k.a. the effect that increasing the value of the independent variable has on the predicted y value)
- ... = do the same for however many independent variables you are testing
- $B_n X_n$ = the regression coefficient of the last independent variable
- ϵ = model error (a.k.a. how much variation there is in our estimate of y)

Tools Utilized





Implementation of Multiple Linear Regression Model

Total Number of Independent Variables
: 62

Dependent Variable/Response Variable
: total_cases (Total confirmed cases of COVID-19. Counts can include probable cases, where reported.)

Location: Canada

Rows: 888

Columns: 62

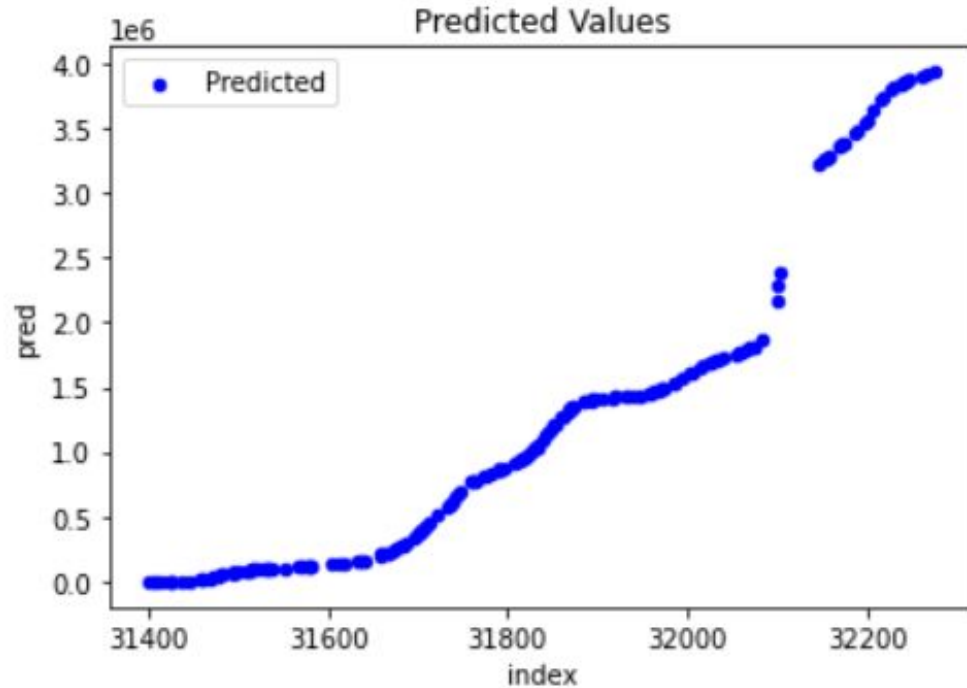
Data and data split info:

size of training data: 710

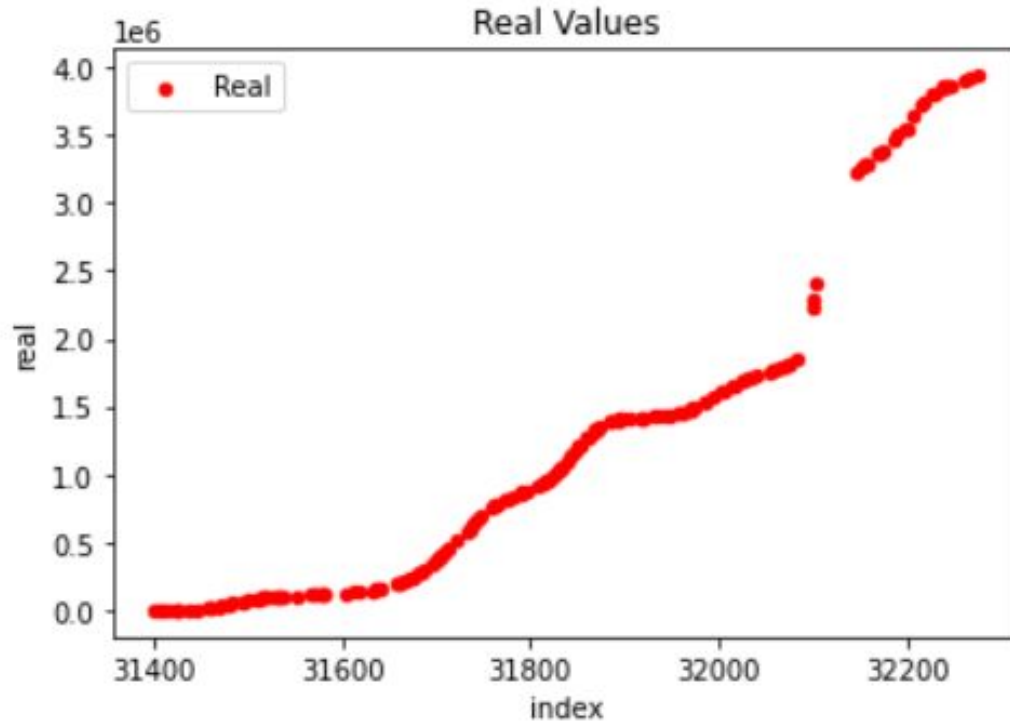
size of testing data: 178

number of features: 62

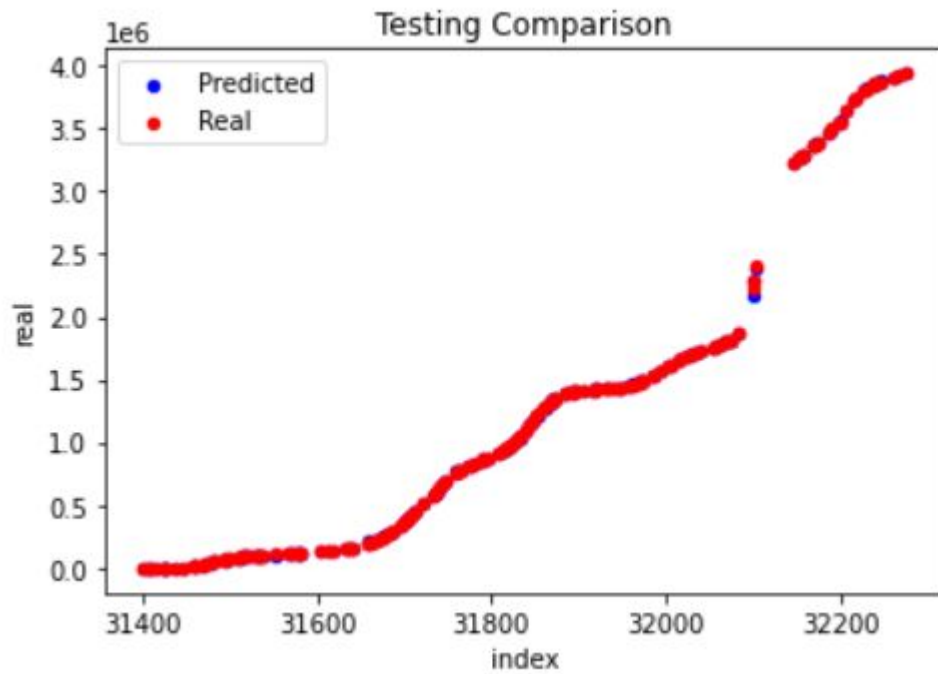
Predicted Values



Real Values



Testing Comparison



Coefficients

coefficients:

	name	value
41	new_people_vaccinated_smoothed_per_hundred	-2516871.112
8	new_cases_smoothed_per_million	-1531184.981
16	hosp_patients_per_million	-548573.616
11	new_deaths_smoothed_per_million	-365578.887
14	icu_patients_per_million	-141536.265
3	total_deaths	-54764.068
37	people_fully_vaccinated_per_hundred	-53607.203
27	positive_rate	-24961.762
38	total_boosters_per_hundred	-22548.508
1	new_cases	-20144.917
4	new_deaths	-7419.666
39	new_vaccinations_smoothed_per_million	-60.204
25	new_tests_smoothed	-33.822
28	tests_per_case	-33.715
42	stringency_index	-33.688
0	index	-31.919
21	total_tests	-7.009
29	total_vaccinations	-0.557
22	new_tests	-0.376
30	people_vaccinated	-0.064
58	excess_mortality_cumulative_absolute	-0.036
33	new_vaccinations	-0.006
20	weekly_hosp_admissions_per_million	-0.005
56	life_expectancy	-0.001
50	cardiovasc_death_rate	-0.001
45	median_age	-0.001

45	median_age	-0.001
53	male_smokers	-0.001
43	population	-0.000
51	diabetes_prevalence	-0.000
47	aged_70_old	-0.000
52	female_smokers	-0.000
44	population_density	-0.000
49	extreme_poverty	-0.000
54	handwashing_facilities	0.000
57	human_development_index	0.000
55	hospital_beds_per_thousand	0.000
46	aged_65_old	0.000
19	weekly_hosp_admissions	0.011
18	weekly_icu_admissions_per_million	0.021
32	total_boosters	0.061
17	weekly_icu_admissions	0.089
31	people_fully_vaccinated	0.141
48	gdp_per_capita	0.481
48	gdp_per_capita	0.481
34	new_vaccinations_smoothed	1.595
61	excess_mortality_cumulative_per_million	2.173
40	new_people_vaccinated_smoothed	6.596
59	excess_mortality_cumulative	16.392
60	excess_mortality	19.601
6	total_cases_per_million	38.051
12	reproduction_rate	3096.940
13	icu_patients	3725.303
5	new_deaths_smoothed	9688.578
24	new_tests_per_thousand	13930.890
15	hosp_patients	14405.930
36	people_vaccinated_per_hundred	24486.056
2	new_cases_smoothed	40225.230
35	total_vaccinations_per_hundred	211903.849
23	total_tests_per_thousand	266837.290
10	new_deaths_per_million	280315.782
7	new_cases_per_million	766879.881
26	new_tests_smoothed_per_thousand	1291330.774
9	total_deaths_per_million	2084763.449



Analysis of Regression

```
Analysis of regression
-----
Score: 0.9999746757931892
R squared: 0.9999726896454993
Mean error: 3057.015806166927 cases
-----
```



Predicted Values

```
[7]: future_y = lr.predict(future_x)

for i in range(days_offset):
    print(f'Predicted cases {i+1} days after {dates.tolist()[-1]}: {int(future_y[-i])}')

Predicted cases 1 days after 2022-06-28: 3931640
Predicted cases 2 days after 2022-06-28: 3935835
Predicted cases 3 days after 2022-06-28: 3932369
```



Conclusion

In Conclusion, the Multiple Linear Regression model provides an almost accurate graph in regards to the number of total cases on a given day. This information would be valuable for policymakers who make severely important decisions on a day to day basis as well as companies that may be affected by COVID. It would allow companies to prepare for surges in the virus and act accordingly.

Future Scope:

- ❖ implementing more than one Machine Learning Model such as Recurrent Neural Networks (RNN)
- ❖ The current model may be improved by replacing NULL values with an average value
- ❖ The Multiple Linear Regression Model could also be tested with other countries as well



THANK YOU!!!